



(Selection & Arrangement)

(Selection)

Permutation & Combination

CONCEPT

- $n! = n(n-1)(n-2)(n-3)\dots \times 3 \times 2 \times 1$
Note: $0!$ (factorial of zero) is 1.
 - ${}^nC_r = n! / [r! \times (n-r)!]$
 - ${}^nP_r = n! / (n-r)!$
 - Number of arrangements of n different objects in a line = $n!$
 - Number of arrangements of n different objects in a circle = $(n-1)!$
 - Number of arrangements of n different objects in a circle where clockwise and anti-clockwise are same = $(n-1)! / 2$
 - Number of arrangements of n objects in a line where p objects are alike of one kind, q are alike of another kind, r are alike of still another kind = $n! / p!q!r!$
 - Number of arrangements when some objects are together = (Arrangement of all objects considering box as single object) $\times$ (Arrangement within box)
- Formulae 7 and 8 are mostly applicable for rearrangement of letters of a word.

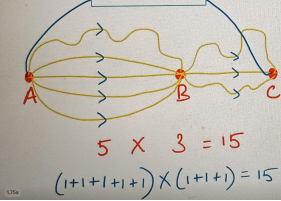
NOTE: For digit problems,

- Draw the dashes equal to number of digits.
- Fill the dashes from left to right, except when some condition is given. For e.g. the condition could be that the units digit is prime or even.
- If Zero is among the given digits, then it cannot occupy leftmost place unless otherwise clarified.

PRINCIPLE OF COUNTING

OR $\Rightarrow +$

AND $\Rightarrow \times$



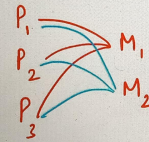
1. You have 3 pens and 2 markers. In how many ways:

i. Can you pick an item (pen or marker)?

Ans: $3 + 2 = 5$

ii. Can you pick a pen and a marker?

Ans: $3 \times 2 = 6$



2. If you have 3 shirts and 2 pairs of pants, in how many different ways can you dress up?

Ans:

$$\begin{array}{ccc} S_1 & P_1 & 1J \\ S_2 & P_2 & J \\ S_3 & & \end{array} \quad \text{(nothing is optional)}$$

$$3 \times 2 = 6$$

$$\begin{array}{ccc} 3S & 2P & 1J \\ S_1 & P_1 & J \\ S_2 & P_2 & J \\ S_3 & & \end{array} \quad \text{(Jacket & shirt is optional)}$$

$$3 \times 2 \times 2 = 12$$

(only Jacket is optional)

3. A shopping mall has 3 distinct glass doors and 2 distinct metal doors for entry, and 5 distinct glass doors and a wooden door for exit.

i. In how many ways can you enter and exit the mall?

Ans: $5 \times 6 = 30$

ii. In how many ways can you enter and exit the mall using only glass doors?

Ans: $3 \times 5 = 15$

iii. In how many ways can you enter and exit the mall without using glass doors?

Ans: $2 \times 1 = 2$

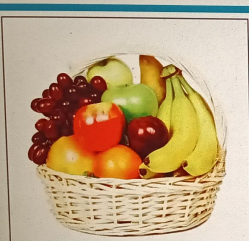
ENTRY

G_1
 G_2
 G_3
 M_1
 M_2

EXIT

G_1
 G_2
 G_3
 G_4
 G_5
 W

(This was only normal counting up till now.)



COMBINATION

- Selection
- Order does not matter

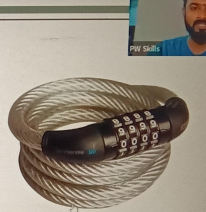
$${}^nC_n = \frac{n!}{n!(n-n)!}$$

$${}^nP_n = \frac{n!}{(n-n)!}$$

$${}^nP_n = {}^nC_n \times n!$$

$$= \frac{n!}{n!(n-n)!} \times n! = \frac{n!}{(n-n)!}$$

$${}^nC_1 = {}^nP_1 = n$$



PERMUTATION

- Selection & Arrangement
- Order matters

7. How many three-digit numbers can be formed so that the unit's place number is prime?

i. When repetition is allowed.

Ans: $9 \times 10 \times 4 = 360$

ii. When repetition is not allowed.

Ans: $8 \times 8 \times 4 = 256$

$$i) \rightarrow \frac{9}{(1-9)} \times \frac{10}{(0-9)} \times \frac{4}{(2,3,5,7)} = 360$$

$$ii) \rightarrow \frac{8}{0} \times \frac{8}{0} \times \frac{4}{(2,3,5,7)} = 256$$

4. How many three-digit numbers can be formed using the digits 7, 8, and 9?

i. When repetition is allowed.

Ans: $3 \times 3 \times 3 = 27$

ii. When repetition is not allowed.

Ans: $3 \times 2 \times 1 = 6$

$$i) \rightarrow \frac{3}{(7,8,9)} \times \frac{3}{(7,8,9)} \times \frac{3}{(7,8,9)} = 27$$

$$ii) \rightarrow \frac{3}{(7,8,9)} \times \frac{2}{(7,8,9)} \times \frac{1}{(7,8,9)} = 6$$

5. How many three-digit numbers can be formed using the digits 0, 7, 8, and 9?

i. When repetition is allowed.

Ans: $3 \times 4 \times 4 = 48$

ii. When repetition is not allowed.

Ans: $3 \times 3 \times 2 = 18$

$$i) \rightarrow \frac{3}{(7,8,9)} \times \frac{4}{(0,7,8,9)} \times \frac{4}{(0,7,8,9)} = 48$$

$$ii) \rightarrow \frac{3}{0} \times \frac{3}{0} \times \frac{2}{(2,3,5,7)} = 18$$

(7,8,9)

9. In how many different ways can the letters of the word "LEADING" be arranged?

i. When repetition is not allowed.

Ans: $7! = 5040$

ii. Such that all the vowels are together.

Ans: $4! \times 3! = 144$

iii. Such that the vowels are together and the consonants are together.

Ans: $2! \times 3! \times 2! = 24$

iv. All the vowels are not together.

Ans: $7! - 144 - 24 = 4872$

$$i) \rightarrow 7 \ 6 \ 5 \ 4 \ 3 \ 2 \ 1 \rightarrow 7! = 5040$$

8. How many four-digit even numbers can be formed using the first 7 whole numbers?

- i. When repetition is allowed. Ans: _____
 ii. When repetition is not allowed. Ans: _____

i) $\rightarrow \frac{6}{(1-6)} \times \frac{7}{(0-6)} \times \frac{7}{(0,2,4,6)} \times \frac{4}{(0,2,4,6)} = 1176$ (0-6)

ii) $\rightarrow \frac{6}{(1-6)} \times \frac{5}{(0)} \times \frac{4}{(0)} \times \frac{1}{(0)} \text{ OR } \frac{5}{(0)} \times \frac{5}{(0)} \times \frac{4}{(0)} \times \frac{3}{(2,4,6)} = 120 + 300 = 420$

10. How many 11-letter words can be formed using all the letters of "MATHEMATICS"?

- i. (No condition.) Ans: _____
 ii. Such that all the vowels are together. Ans: _____

M A T H E M A T I C S

i) $\frac{11!}{2! \times 2! \times 2!}$

ii) $\frac{8!}{2! \times 2!} \times \frac{4!}{2!}$

i) $[E, A, I], [L, D, N, G] \Rightarrow 5! \times 3! = 120 \times 6 = 720$
 ii) $[E, A, I], [L, D, N, G] \Rightarrow 2! \times 3! \times 4! = 2 \times 6 \times 24 = 288$
 iii) $n(\text{Vowels NOT together}) = n(\text{TOTAL}) - n(\text{Together}) = 5040 - 720 = 4320$

11. Loki has 1 red and 3 green identical marbles. In how many ways can he arrange them in a straight line?

Ans: _____

$R G G G \mid \frac{4!}{3!} = 4$

12. A committee of 10 people needs to be seated on 10 chairs in a straight line. In how many different ways can they be seated if 8 particular people never sit together?

Ans: _____

$n(\text{Total}) = 10!$

$n(\text{Together}) = \frac{10!}{3! \times 8!}$

$n(\text{not together}) = 10! - 3! \times 8!$

total people - the 8 people who sat together (as we can't directly find the people who didn't sit together).

13. In how many ways can we select a team of 4 men and 2 women from a group of 8 men and 5 women?

Ans: _____

${}^8C_4 \times {}^5C_2$

$\frac{8!}{4! \times 4!} \times \frac{5!}{2! \times 3!}$

$\frac{8 \times 7 \times 6 \times 5}{4 \times 3 \times 2} \times \frac{5 \times 4}{2} = 700$

16. There are 10 people in a meeting. Everybody shakes hand with everybody. What is the minimum number of handshakes possible?

Ans: _____

$\frac{10 \times 9}{2} \text{ OR } \frac{10C2}{2}$

14. In how many ways can a team of 11 be selected from 5 men and 11 women such that the team must comprise of not more than 3 men?

- A) 1565 B) 2256 C) 2255 D) 1243

$3M \& 8W \text{ OR } 2M \& 9W \text{ OR } 1M \& 10W \text{ OR } 11W$
 ${}^5C_3 \times {}^{11}C_8 + {}^5C_2 \times {}^{11}C_9 + {}^5C_1 \times {}^{11}C_{10} + {}^{11}C_{11}$

$\frac{5!}{3! \times 2!} \times \frac{11!}{8! \times 3!} + \frac{5!}{2! \times 3!} \times \frac{11!}{9! \times 2!} + 5 \times 11 + 1$

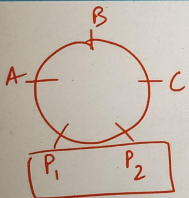
$\frac{5 \times 4}{2} \times \frac{11 \times 10 \times 9}{3 \times 2} + \frac{5 \times 4}{2} \times \frac{11 \times 10}{2} + 55 + 1$

$1650 + 550 + 55 + 1$

$= 2256$

18. In how many ways can 5 family members sit around a circle such that parents always sit together?

Ans: _____



$(4-1)! \times 2!$

$3! \times 2!$

$6 \times 2 = 12$

$\rightarrow$ How many people can sit in a line = $n!$
 but for a circle it's $(n-1)!$

19. How many necklaces/garlands can be formed with 7 beads/flowers?

Ans: _____

$n - n!$

$n - (n-1)!$

$n - \frac{(n-1)!}{2} = \frac{(7-1)!}{2} = \frac{6!}{2} = 360$

$\div 2$ Kiya isme Kyuki a necklace is symmetrical from the left & right.

20. Out of 7 consonants and 4 vowels, how many words of 3 consonants and 2 vowels can be formed?

A) 24400

B) 21300

C) 210

✓ D) 25200

$$\text{Selection} = {}^7C_3 \times {}^4C_2 = \frac{7!}{3! \times 4!} \times \frac{4!}{2! \times 2!}$$
$$= \frac{7 \times 6 \times 5 \times 4}{2 \times 2} = 210$$

$$\text{Selection \& Arrangement} = 210 \times 5!$$
$$= 210 \times 120 = \underline{\underline{25200}}$$

or

$$\underline{\underline{{}^7P_3 \times {}^4P_2 = 25200}}$$

